

Home Lab — Day 1 & 1b: Active Directory + Domain-Joined Client

Self-directed home lab to practise Level 1 service desk / IT support tasks. Not paid or production work — built for learning and interview preparation.

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Environment

Host	Windows host, 16 GB RAM, Oracle VirtualBox 7.2
DC01 (server)	Windows Server 2022 Standard (Eval), Desktop Experience — 4 GB RAM, 2 vCPU, 60 GB. Static IP 10.0.2.15/24, DNS → self. Domain Controller for forest corp.lab.local (NetBIOS CORP).
CLIENT01 (client)	Windows 11 Pro — 4 GB RAM, 2 vCPU, 60 GB, EFI + TPM 2.0. Local admin 'labadmin'. Static IP 10.0.2.20, DNS → 10.0.2.15. Joined to corp.lab.local.
Networking	Both VMs on a VirtualBox Internal Network with static IPs (isolated lab, no internet) — sufficient for domain authentication and policy testing.

Tasks completed

Task	What I did	Why it matters (L1 relevance)
Build Domain Controller	Installed Server 2022 from scratch, set static IP + self-DNS, renamed to DC01, installed AD DS, promoted to DC in new forest corp.lab.local.	The core directory every service desk works against.
OU structure	Built CORP > Users, Groups, Computers, Departments > IT, Sales.	Organises objects and targets Group Policy; real-world practice.
User creation	Created alex.taylor in IT OU (firstname.lastname, must-change-at-next-logon).	The standard 'new starter account' ticket.
Password reset	Reset password with temp + force-change + unlock options.	The #1 L1 ticket.
Lockout policy	Set account lockout threshold to 3 via Default Domain Policy.	Makes lockouts realistic and observable.
Security group	Created Global Security group IT-Staff; added the user.	Access via group membership, not per-user.
Domain join	Joined Windows 11 client to corp.lab.local using CORP\Administrator credentials.	Setting up a new starter's PC; core sysadmin task.
Live lockout loop	Logged in as the user (forced password change), triggered a real lockout with 3 bad attempts, unlocked from DC01 in ADUC, logged back in.	The full 'I'm locked out' workflow, end to end.
Group Policy	Created/linked a 'Logon Banner' GPO (interactive logon message), applied with gpupdate /force, confirmed banner on the client.	Central policy applied to a managed machine — core GPO skill.

Troubleshooting worked through (real problems, real fixes)

- **UEFI black screen:** empty disk had no EFI bootloader, so firmware wouldn't fall through to the DVD — fixed by adjusting boot order / forcing the optical device.
- **Win 11 requirements:** TPM 2.0 + Secure Boot needed; re-enabled EFI and set TPM 2.0 in VirtualBox.
- **Oobe crash + forced Microsoft account:** resolved by disconnecting the network to unlock local-account setup.
- **IP address clash / DNS:** client mistakenly shared DC01's IP; corrected to a unique static IP (10.0.2.20) with DNS pointed at the DC.
- **VM-to-VM connectivity:** switched both VMs to an Internal Network with static IPs so they could see each other reliably.

Concepts learned

Why a DC needs a static IP and self-DNS • UEFI vs legacy BIOS boot • domain authentication (client validates against the DC, not locally) • Group Policy applies automatically at boot/logon and refreshes ~every 90 min; `gpupdate /force` or `Invoke-GPUUpdate -Computer` forces it without user involvement.

Next: Day 2 — Microsoft 365 Developer tenant + Entra ID, repeating the same identity tasks in the cloud. Evidence: screenshots captured at each milestone, for a self-directed home-lab GitHub repository.